
Intelligent Network Intrusion Detection Using KNN and Feature Scaling for Enhanced Cyber Security Analysis.

Problem ID: 09
Team ID: PY-GT-58

Name : Sambhav Jaiswal(Team Leader)

Class Roll Number: 58

University Roll Number : 2028835

Section and Group: G(G-2)

Name : Pragun Khanna

Class Roll Number:54

University Roll Number : 2028798

Section and Group: G(G-2)

Name : Utsav Bajoria

Class Roll Number:73

University Roll Number : 2208896

Section and Group: G(G-2)

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1. Problem Description

In the modern digital world, cybersecurity is a major concern due to the rapid increase in cyber attacks. Networks are constantly exposed to threats such as hacking, malware, and unauthorized access. Detecting these threats manually is difficult because of the large amount of data generated in network traffic.

This project aims to build a basic Intrusion Detection System (IDS) using Machine Learning techniques. The system will analyse network data based on features like Packet Size, Connection Duration, and Error Rate. These features help in identifying patterns that indicate whether the traffic is normal or suspicious. The K-Nearest Neighbours (KNN) algorithm will be used for classification. It is simple and suitable for beginners as it classifies data based on similarity. The Elbow Method will be used to find the optimal value of K, which improves the model's accuracy. This project helps first-year students understand how machine learning can be applied in real-world cybersecurity problems.

2. Architecture



Fig. 1 Sample architecture

3. Models Used(Applied/ Planned)

Give a brief introduction of the models used:

1. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a supervised machine learning classification algorithm used to classify data points based on similarity with neighboring data points. It identifies the K closest data samples and assigns the majority class among them. In this project, KNN is the primary model for intrusion detection because it is simple, beginner-friendly, and effective for detecting suspicious network traffic patterns after feature scaling.

2. Logistic Regression

Logistic Regression is a supervised classification algorithm used for binary classification problems such as Normal vs Intrusion detection. It predicts the probability of a data point belonging to a particular class using a logistic (sigmoid) function. In this project, Logistic Regression is used as a baseline model for comparing performance with other advanced algorithms.

3. Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm that classifies data by finding the optimal hyperplane that best separates different classes. It is effective for both linear and nonlinear classification problems. In this project, SVM is used to improve classification accuracy for complex intrusion patterns in network traffic.

4. Decision Tree

Decision Tree is a supervised learning algorithm that makes decisions by splitting data into branches based on feature conditions. It works like a flowchart where each node represents a decision rule. In this project, Decision Tree is used because it is easy to interpret and helps identify which network features contribute most to intrusion detection.

5. Random Forest

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting. It gives more reliable predictions by averaging the outputs of several trees. In this project, Random Forest is used for advanced performance comparison and is expected to provide strong accuracy in intrusion detection.

4. Evaluation metrics used

Accuracy:

Measures the overall percentage of correct predictions made by the model.

Precision:

Measures how many predicted intrusion cases were actually correct.

Recall:

Measures how many actual intrusion cases were successfully detected.

F1-Score:

Provides a balanced measure of Precision and Recall.

Confusion Matrix:

Shows the number of correct and incorrect predictions, including false positives and false negatives.

Classification Report:

Provides a complete summary of Precision, Recall, F1-Score, and Support for each class.

5. Preliminary Results and Visualizations

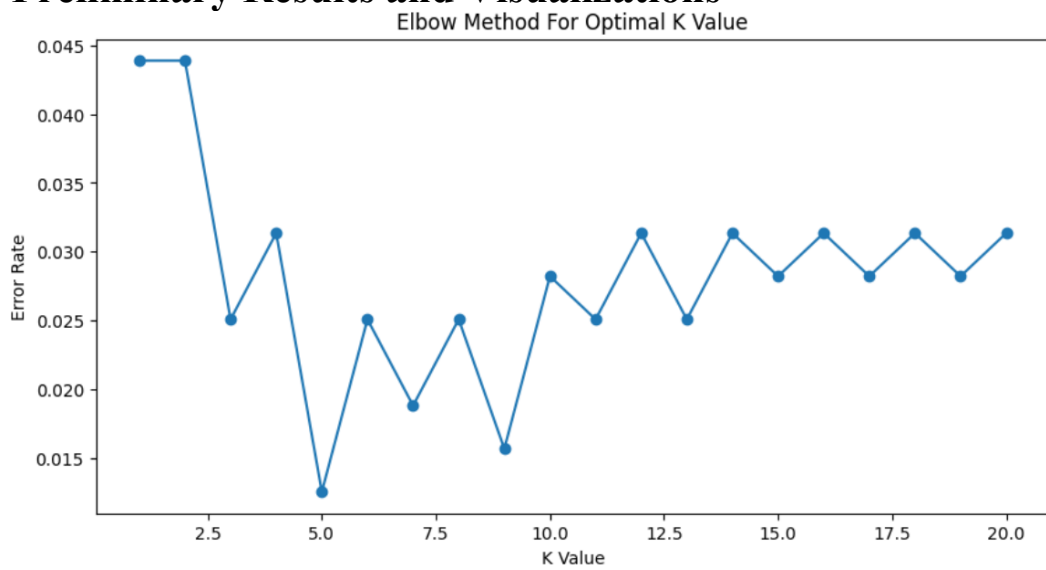


Fig.1: Showing graph for elbow method showing optimal K value

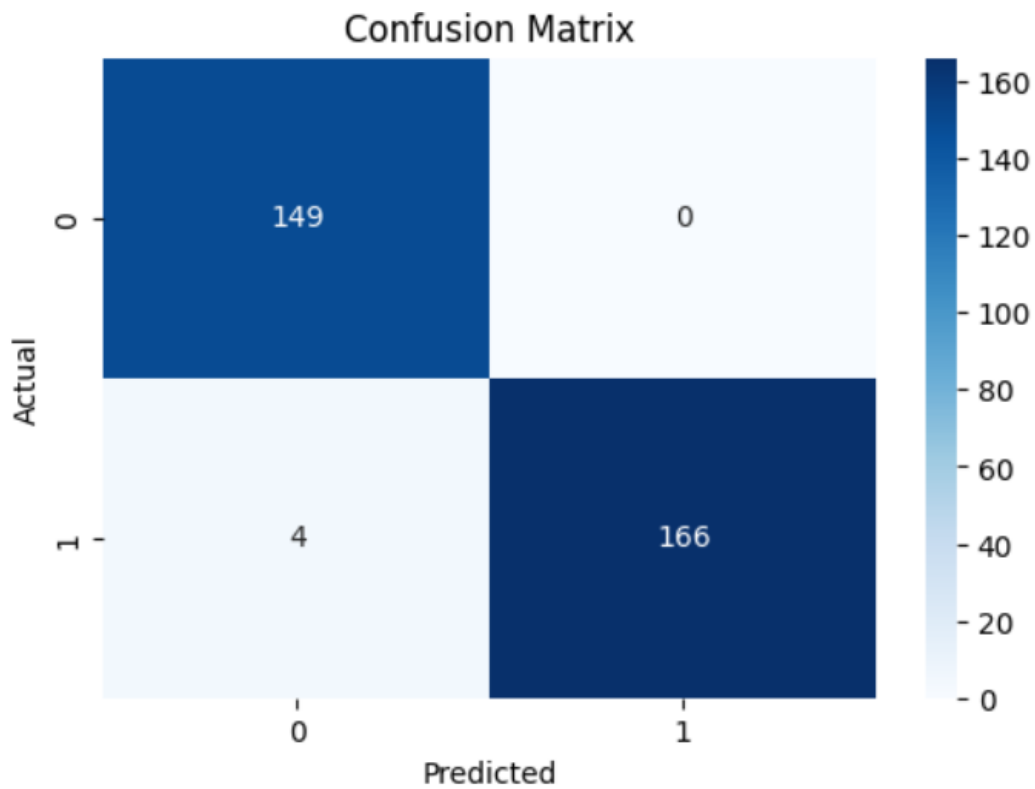


Fig 2. Graph of Confusion Matrix showing the actual value and predicted value.

LOGISTIC REGRESSION

Accuracy: 99.05956112852664 %

Classification Report:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	149
1	1.00	0.98	0.99	170
accuracy			0.99	319
macro avg	0.99	0.99	0.99	319
weighted avg	0.99	0.99	0.99	319

Confusion Matrix:

```
[[149  0]
 [ 3 167]]
```

SUPPORT VECTOR MACHINE

Accuracy: 99.37304075235109 %

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	149
1	1.00	0.99	0.99	170
accuracy			0.99	319
macro avg	0.99	0.99	0.99	319
weighted avg	0.99	0.99	0.99	319

Confusion Matrix:

```
[[149  0]
 [ 2 168]]
```

Fig 3

RANDOM FOREST

Accuracy: 100.0 %

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	149
1	1.00	1.00	1.00	170
accuracy			1.00	319
macro avg	1.00	1.00	1.00	319
weighted avg	1.00	1.00	1.00	319

Confusion Matrix:

```
[[149  0]
 [  0 170]]
```

DECISION TREE

Accuracy: 100.0 %

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	149
1	1.00	1.00	1.00	170
accuracy			1.00	319
macro avg	1.00	1.00	1.00	319
weighted avg	1.00	1.00	1.00	319

Confusion Matrix:

```
[[149  0]
 [  0 170]]
```

Fig 4

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
- Stack Overflow – <https://stackoverflow.com>